

Vaishak Balachandra

📍 United States ✉ vaishakbalachandra01252002@gmail.com ☎ 2193682467 📺 in/vaishakbalachandra 🌐 sites.google.com/view/vaishakbalachandra/

SUMMARY

Machine Learning Engineer with 3.82 GPA in MS Computer Science and 1 year of LLM production experience at Boston Limited. Expertise in RAG Architectures, LLM Fine-Tuning, and agentic workflows using Python, PyTorch, and TensorFlow. AWS and GCP ML Specialty certified with proven ability to build, test, and deploy GenAI applications at scale.

EXPERIENCE

Graduate Research Assistant

Purdue University Northwest

August 2024 - Present, Hammond, IN

- Conducting original thesis research on VertiSplitRAG, a novel privacy-preserving RAG architecture using vertical attribute partitioning, targeting 95%+ retrieval accuracy with 40% latency reduction across 10,000+ query evaluations.
- Designed end-to-end evaluation pipelines measuring retrieval accuracy across 5 benchmark datasets and 10,000+ queries, quantifying differential privacy preservation with epsilon values from 0.1 to 1.0.
- Benchmarked LLM optimization techniques including FlashAttention-2, INT8/FP16 quantization, and mixed-precision inference across GPT-2 and Mistral-7B on NVIDIA A100, producing open-sourced technical reports and Jupyter notebooks.

Machine Learning Engineer

Boston Limited

November 2023 - December 2024, Bengaluru, India

- Executed end-to-end model development, fine-tuning and regression testing of GenAI applications, improving inference accuracy and accelerating defect resolution. Led nightly regressions and pioneered failure mode severity analysis across LLaMA-2 deployments, ensuring faster issue resolution.
- Developed and automated RAG pipeline tests using Python and LangChain, reducing manual effort and executing 100+ functional and integration tests, ensuring a 100% on-time product release rate. Monitored BLEU, ROUGE and other scores to validate model performance.
- Led development of critical GenAI features, including dense passage retrieval, instruction fine-tuning (SFT), and RLHF-based streamlining model, reducing deployment timelines by 30%, and identifying critical hallucination failure points for improved output reliability.

PROJECT

VertiSplitRAG: Privacy-Preserving Retrieval-Augmented Generation

Purdue University Northwest • August 2024 - Present

- Designed a novel RAG architecture using vertical attribute partitioning to isolate sensitive user data across distributed retrieval nodes, ensuring no single node holds sufficient context to reconstruct user profiles, achieving 95%+ retrieval accuracy while maintaining differential privacy guarantees.
- Implemented end-to-end evaluation pipelines across 5 benchmark datasets measuring retrieval accuracy on 10,000+ queries, analyzing latency tradeoffs (40% reduction achieved), and quantifying privacy preservation metrics with epsilon values ranging from 0.1 to 1.0.
- Developed distributed retrieval system using Python, LangChain, and FAISS vector database, with comprehensive documentation and reproducible experiments planned for IEEE Smartnets submission.

LLMShield: Adversarial Attack Simulation and Mitigation in LLM Pipelines

Purdue University Northwest • github.com/vaishakbalachandra/llm-inference-optimization • January 2025 - June 2025

- Simulated prompt injection and jailbreak attacks using custom adversarial datasets containing 10,000+ adversarial prompts, identifying critical vulnerabilities in LLaMA-2 and GPT-4 inference pipelines across RAG-based systems.
- Conducted high-volume stress testing on RAG architectures, analyzing failure modes across context poisoning, embedding inversion, and retrieval manipulation scenarios, improving output reliability by 35% through targeted mitigation strategies.
- Implemented security testing framework using LangChain and NVIDIA NeMo Guardrails, developing automated detection and prevention mechanisms for common LLM attack vectors in production environments.

EDUCATION

Master of Science in Computer Science Engineering

Purdue University Northwest, Hammond • Hammond, IN • 2026 • 3.82 GPA

- Thesis Track

Bachelor of Engineering in Electronics and Telecommunication Engineering

RV College of Engineering, Bengaluru • Bengaluru, KA • 2024 • 8.55 CGPA

- 6th Rank in Department, GATE Qualified

CERTIFICATIONS

Deep Learning Specialization

DeepLearning.AI • 2025

- Comprehensive training in neural networks, CNNs, RNNs, and optimization techniques, foundational for LLM fine-tuning and transformer-based model development.

AWS Certified Machine Learning - Specialty

Amazon Web Services (AWS) • 2024

- Validates expertise in building, training, and deploying machine learning models on AWS cloud infrastructure, directly applicable to production ML engineering roles.

Google Cloud Professional Machine Learning Engineer

Google Cloud • 2024

- Demonstrates proficiency in designing, building, and productionizing ML models using Google Cloud AI/ML services and multi-cloud deployment strategies.

Building RAG Agents with LLMs

NVIDIA Deep Learning Institute • 2024

- Specialized training in Retrieval-Augmented Generation architectures, directly aligned with thesis research and production RAG pipeline development experience.

INVOLVEMENT

Treasurer

Purdue University Northwest, Hammond • Indian Student Association at Purdue Northwest (ISAP) • January 2025 - February 2026

- Managed organization budget and financial planning for cultural events serving 400+ students.
- Coordinated financial planning and budget allocation for 10+ cultural events throughout the academic year, managing vendor payments and expense tracking for programs serving 200+ community members.
- Collaborated with executive board to develop and implement transparent financial reporting processes, ensuring fiscal accountability and supporting organization's achievement of Program of the Year award for two consecutive years.

PUBLICATIONS

VertiSplitRAG: Privacy-Preserving Retrieval-Augmented Generation Through Vertical Attribute Partitioning

Conference Paper • IEEE Smartnets • 2026

- Submitted for review. Novel RAG architecture using vertical attribute partitioning for privacy preservation across distributed retrieval nodes.

Fake Currency Detection using ML and Image Processing: An Application for Blind People using Android Studio

Conference Paper • IEEE ICACRS • 2023

- AI for Good Award-winning paper. Developed ML-powered counterfeit detection system achieving 95% accuracy with accessibility features for visually impaired users.

PneumoGAN: GAN Based Pneumonia Detection using Synthetic Data Generation

Open Source / Preprint • GitHub / arXiv • 2024

- Open-source contribution. GAN-based synthetic medical image generation achieving 94.99% pneumonia detection accuracy on NIH chest X-ray dataset.

SKILLS

Python, PyTorch, TensorFlow, Keras, HuggingFace Transformers, LangChain, LlamaIndex, Scikit-learn, RAG Architectures, LLM Fine-Tuning, RLHF, Agentic Workflows, Multi-Agent Systems, Prompt Engineering, BLEU, ROUGE, Perplexity Scoring, A/B Testing, Hallucination Analysis, Regression Testing, LLM-as-a-Judge, CI/CD Pipelines, MLflow, Docker, Kubernetes, vLLM, Model Monitoring, Apache Airflow, AWS SageMaker, Google Cloud Vertex AI, IBM Watson Studio, Multi-Cloud Deployment, SQL, PostgreSQL, BigQuery, Spark, R, Power BI, Jupyter Notebooks, Code Reviews, Debugging, Technical Documentation, JIRA, Git, FlashAttention-2, INT8 Quantization, FP16 Quantization, NVIDIA A100, Dense Passage Retrieval, Instruction Fine-Tuning, Differential Privacy, Distributed Systems